CLOUD SECURITY — UNIT 4 ASSIGNMENT

Launch a Windows VM and Connect Using RDP

Step-by-step practical report with remote access configuration, PowerShell commands, representative output, and security checks

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Execution and output note:
This report contains commands and representative expected outputs for documentation. The output examples are illustrative and may differ by operating-system image, package version, region, or account settings. They are not represented as captured execution evidence.

1. AIM

To launch a Windows Server virtual machine on Amazon EC2, restrict Remote Desktop Protocol (RDP) access to the user's IP address, connect using RDP, inspect system information, create a local user, verify Windows Firewall status, and create a test text file.

2. REQUIREMENTS

- An AWS account with permission to create and manage EC2 instances and security groups.
- A Windows Server EC2 AMI and a key pair selected at launch.
- A Windows client with Remote Desktop Connection (`mstsc.exe`), or a compatible RDP client.
- The EC2 administrator password retrieved through the AWS console using the selected key pair; do not include passwords in the report.

3. SECURITY GROUP CONFIGURATION

Inbound rule	Protocol / Port & Source
RDP	TCP 3389 | Source: My IP (your current public IPv4 address /32)
HTTP / SSH	Not required for this RDP-only task unless separately needed
Outbound	Default outbound rules are acceptable for a basic lab unless the instructor specifies otherwise

Critical security note: Do not expose RDP (3389) to `0.0.0.0/0`. Limit access to your own public IP and remove or tighten the rule when the practical is finished.

4. PROCEDURE

Step 1 — Launch a Windows Server EC2 instance

- Open AWS Management Console → EC2 → Instances → Launch instances.
- Choose a Windows Server AMI, select an instance type appropriate for the lab, and choose or create a key pair.
- Create or select a security group with TCP 3389 allowed only from My IP.
- Launch the instance and wait for Running status and successful status checks.

Step 2 — Retrieve the administrator password and connect

Select the instance in the EC2 console and use **Connect** → **RDP client** → **Get password**. Use the private key associated with the instance to decrypt the password. Keep the password private; do not paste it into this report. Download the Remote Desktop file if provided, or open Remote Desktop Connection and enter the instance's public IPv4 address or public DNS name.

```
Remote Desktop Connection (mstsc.exe) Computer: <EC2-PUBLIC-IP-OR-DNS> User name: Administrator Password: <enter privately; do not record in report> Representative connection result: Remote Desktop session opens to the Windows Server desktop. The Windows desktop and Server Manager are visible.
```

Step 3 — Check system information in PowerShell

Open PowerShell as Administrator on the remote Windows Server and run:

```
Get-ComputerInfo | Select-Object CsName, WindowsProductName, WindowsVersion, OsArchitecture, CsTotalPhysicalMemory
```

Representative output format (values vary by AMI and instance type):

```
CsName : EC2AMAZ-EXAMPLE WindowsProductName : Windows Server WindowsVersion : <version varies> OsArchitecture : 64-bit CsTotalPhysicalMemory : <memory varies by instance type>
```

Step 4 — Create a local user account

Run PowerShell as Administrator. The command prompts for a password without printing it. Use a strong password that meets the server's password policy. This example creates a standard local account and does not add it to the Administrators group.

```
$Password = Read-Host "Enter a strong password for LabUser" -AsSecureString New-LocalUser -Name "LabUser" -Password $Password -FullName "Cloud Security Lab User" -Description "Created for Unit 4 practical" Get-LocalUser -Name "LabUser" | Select-Object Name, Enabled, Description
```

```
Name Enabled Description ---- ------- ----------- LabUser True Created for Unit 4 practical
```

Step 5 — Enable and verify Windows Firewall

```
Set-NetFirewallProfile -Profile Domain, Private, Public -Enabled True Get-NetFirewallProfile | Select-Object Name, Enabled, DefaultInboundAction, DefaultOutboundAction
```

```
Name Enabled DefaultInboundAction DefaultOutboundAction ---- ------- ------------------ -- -------------------- Domain True NotConfigured NotConfigured Private True NotConfigured NotConfigured Public True NotConfigured NotConfigured
```

The Enabled field should be True for each profile. Default action values can differ depending on the image and local policy.

Step 6 — Create and verify a text file

```
New-Item -Path C:\CloudSecurityUnit4.txt -ItemType File -Force Set-Content -Path C:\CloudSecurityUnit4.txt -Value "Unit 4 Windows EC2 RDP practical" Get-Content C:\CloudSecurityUnit4.txt Get-Item C:\CloudSecurityUnit4.txt | Select-Object FullName, Length, LastWriteTime
```

```
Unit 4 Windows EC2 RDP practical FullName Length LastWriteTime -------- ------ -------- ----- C:\CloudSecurityUnit4.txt 35 <date/time varies>
```

5. SECURITY AND VALIDATION CHECKLIST

- The Windows EC2 instance is running and status checks pass.
- RDP inbound access is limited to the current public IP (/32).
- RDP login succeeds using the correct instance administrator credentials.
- PowerShell returns Windows system information.
- The `LabUser` account is present and enabled; its password is not documented.

6. TROUBLESHOOTING

Issue	Cause & Resolution
RDP connection times out	Check instance state, public IP, subnet routing, Windows RDP availability, and inbound TCP 3389 rule.
Unable to decrypt password	Confirm that the private key matches the key pair selected when the instance was launched.
Incorrect credentials	Use the correct administrator username and decrypted password; check keyboard layout.
PowerShell command not found	Run on the Windows Server instance in an elevated PowerShell session; some commands require appropriate privileges.

7. CLEAN-UP

When the faculty no longer needs the instance, stop or terminate it according to the lab instructions. Review attached EBS volumes, snapshots, and other billable resources. Keep RDP restricted throughout and do not leave a broad inbound rule in place.

8. CONCLUSION

This practical covers deployment and remote administration of a Windows Server EC2 instance through RDP. It demonstrates restricted network access, retrieval and private handling of connection credentials, system-information inspection, local user creation, firewall verification, and basic file creation.

9. REFERENCES

- Amazon EC2 Windows instances:
  `https://docs.aws.amazon.com/AWSEC2/latest/WindowsGuide/EC2_GetStarted.html`
- Connect to a Windows instance:
  `https://docs.aws.amazon.com/AWSEC2/latest/WindowsGuide/connecting_to_windows_instance.html`
- Amazon EC2 security groups:
  `https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/ec2-security-groups.html`
- Microsoft Learn — Get-ComputerInfo:
  `https://learn.microsoft.com/powershell/module/microsoft.powershell.management/get-computerinfo`
- Microsoft Learn — Windows Firewall with Advanced Security:
  `https://learn.microsoft.com/windows/security/operating-system-security/network-security/windows-firewall/`

UNIT 4 SCREENSHOT APPENDIX

01 Adobe security.txt

Browser displaying Adobe security.txt. Supplementary security reference; not evidence of Windows RDP.

02 Firewall port filter

```
PS C:\Temp> Get-NetFirewallPortFilter -All Protocol : ICMPv6 | LocalPort : Any |
RemotePort : Any Protocol : TCP | LocalPort : RPC | RemotePort : Any Protocol : ICMPv4 |
LocalPort : Any | RemotePort : Any Protocol : TCP | LocalPort : 135 | RemotePort : Any
```

PowerShell output of Get-NetFirewallPortFilter -All. Shows firewall port-filter information.

03 Local user creation

```
PS C:\> New-LocalUser -Name "JohnDoe" -Description "Test User Account" -Password
(ConvertTo-SecureString "Pass@word1" -AsPlainText -Force) Name Enabled Description
JohnDoe True Test User Account
```

PowerShell output showing creation and listing of a local user. Example password visible in original image has been redacted in this report.

04 Computer model inventory

```
PS C:\> $res.Format-Table -AutoSize PSComputerName ComputerModel 2012R2 Microsoft
Corporation Virtual Machine SERVER2016 Microsoft Corporation Virtual Machine WIN7PSV3
Microsoft Corporation Virtual Machine STUEPC System manufacturer System Product Name
DELL2010 Dell Inc. Latitude E6420 ELITEBOOK2015 Hewlett-Packard HP EliteBook 840 G2
```

PowerShell inventory output listing computer names and models. This is not the requested Get-ComputerInfo output from the target EC2 instance.

UNIT 4 SCREENSHOT APPENDIX (CONTINUED)

05 Get-ComputerInfo

```
      TypeName: Microsoft.PowerShell.Commands.ComputerInfo Name MemberType Definition --
-- ---------- ---------- Equals Method bool Equals(System.Object obj) GetHashCode Method
int GetHashCode() GetType Method type GetType() ToString Method string ToString()
BiosBIOSVersion Property string[] BiosBIOSVersion {get;} BiosManufacturer Property string
BiosManufacturer {get;}
```

PowerShell Get-ComputerInfo command. The displayed output is the property list, not a concise Windows Server identity summary.

06 AWS EC2 search

AWS console search for EC2 service. It does not show a running Windows instance.

07 Inbound rule port 8080

Azure network security rule configuration for port 8080, not AWS RDP port 3389.